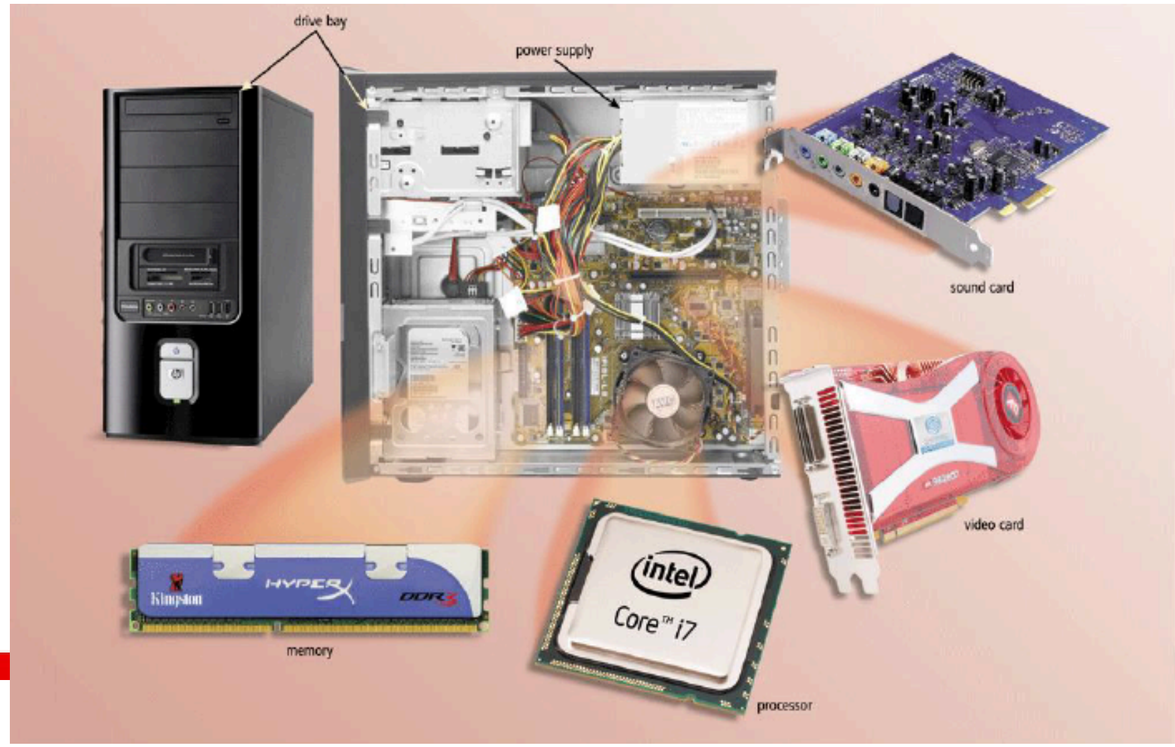
HARDWARE

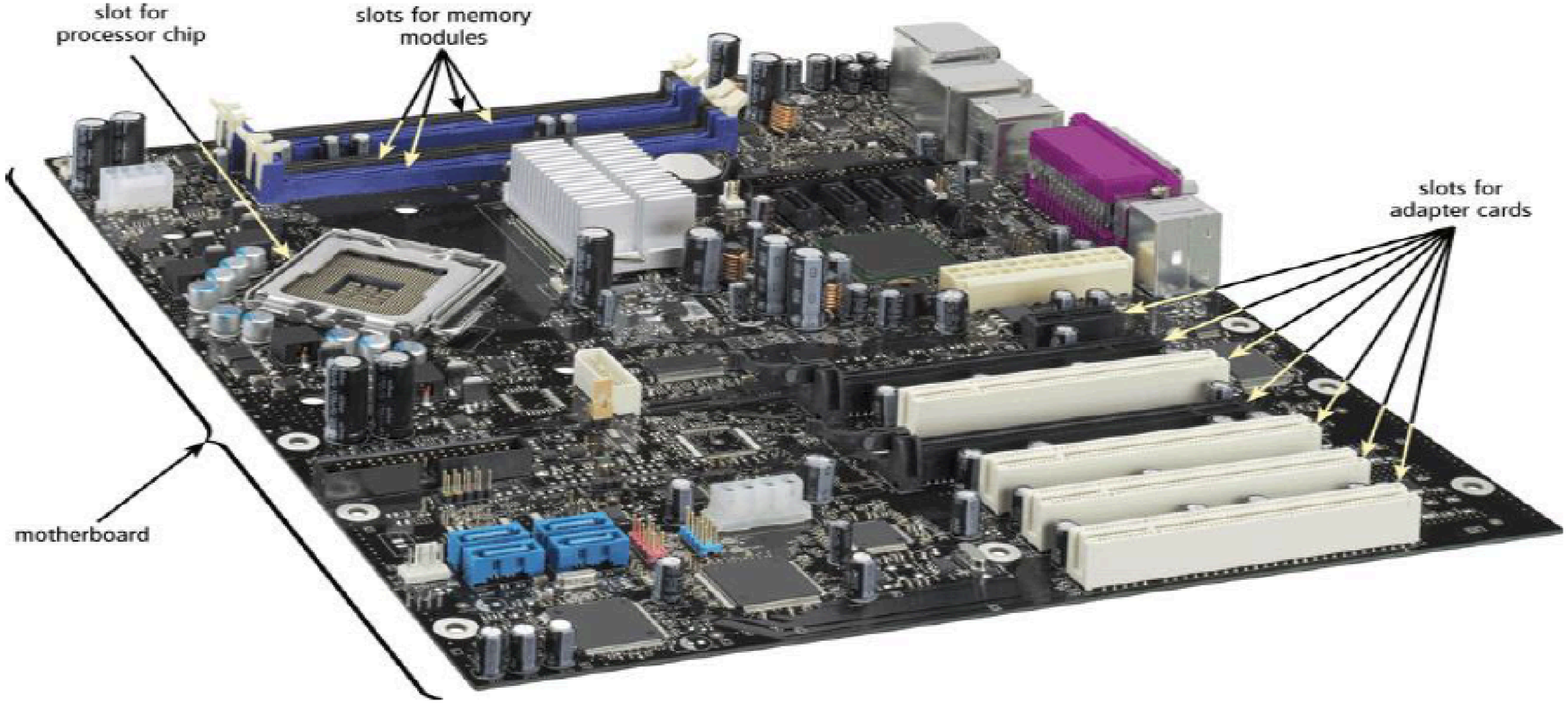
THE SYSTEM UNIT

- The System Unit is a case that contains the electronic components of the computer used to process data
- The inside of the system unit on a desktop PC includes the following:
  - Drive bay(s)
  - Power supply
  - Sound card
  - Video card
  - Processor
  - Memory



THE SYSTEM UNIT

- The motherboard is the main circuit board of the system unit

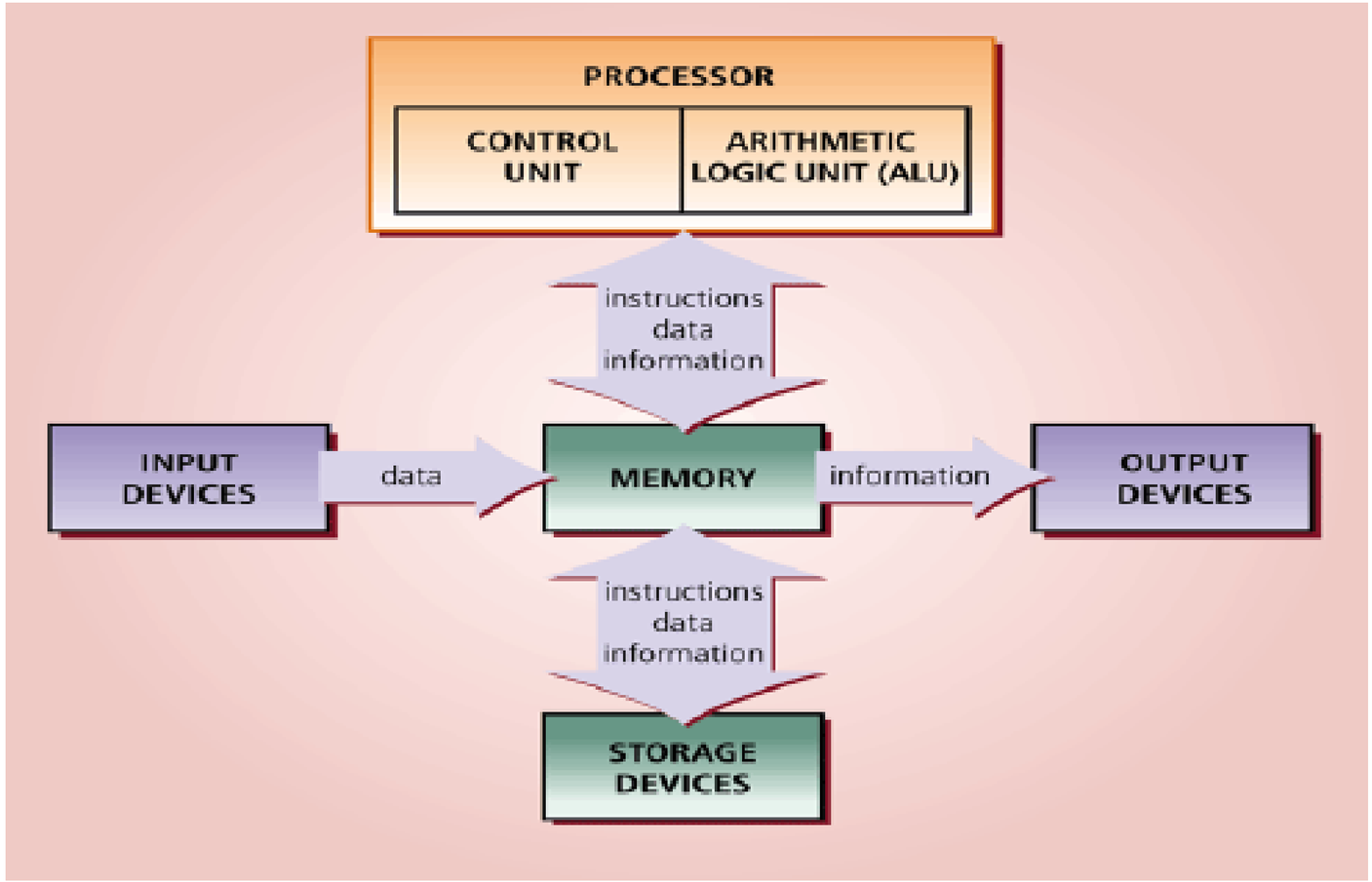


PROCESSOR

- The processor, also called the Central Processing Unit (CPU), interprets and carries out the basic instructions that operate a computer.
- It also contains a Control Unit and an Arithmetic Logic Unit (ALU)

Multi-core processor	Dual-core processor	Quad-core processor	Octa-core processor	Deca-core processor (Future) Intel i9

PROCESSOR

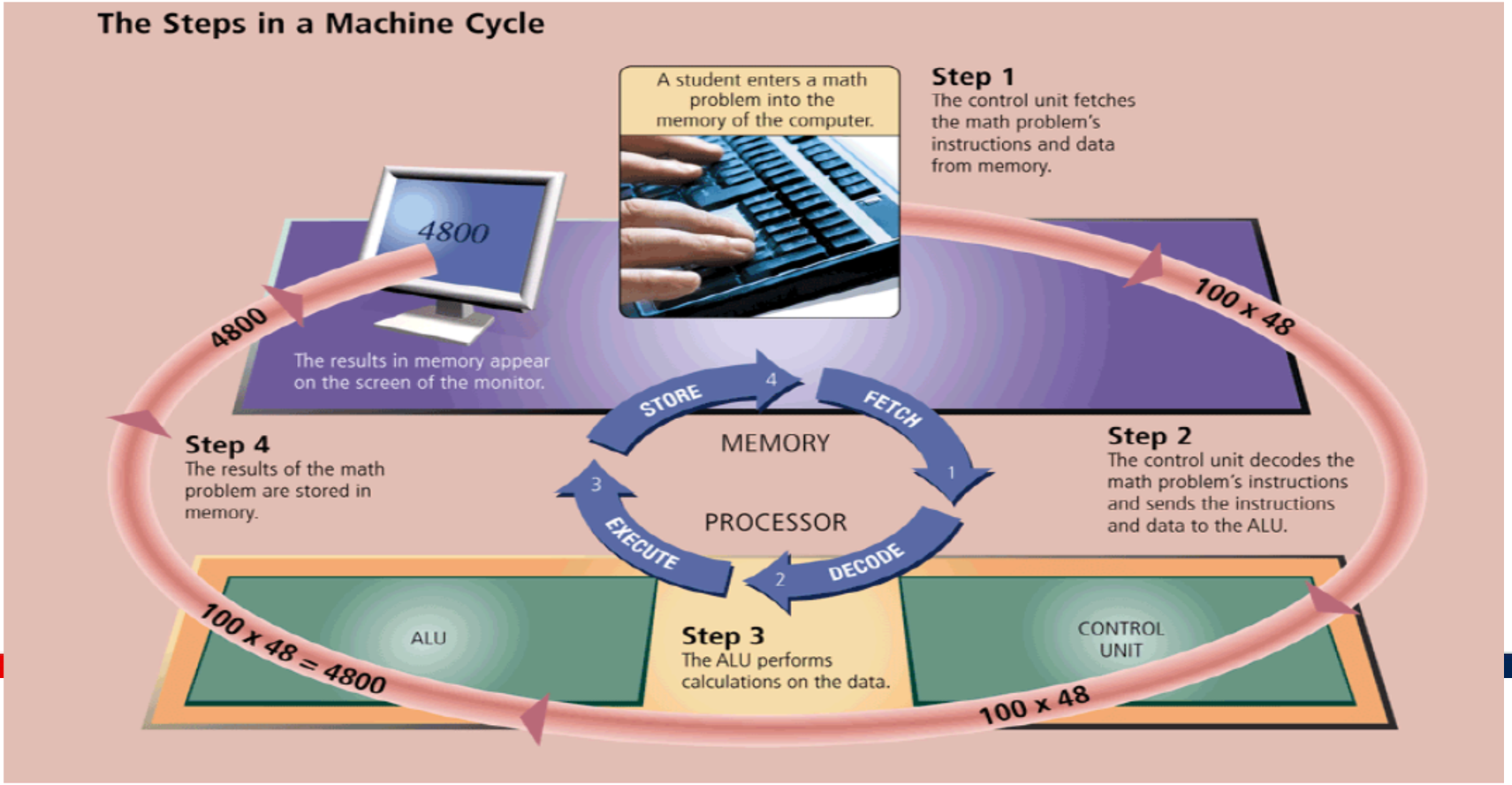

PROCESSOR
CONTROL UNIT
ARITHMETIC LOGIC UNIT (ALU)
instructions
data
information
INPUT DEVICES
data
MEMORY
information
OUTPUT DEVICES
instructions
data
information
STORAGE DEVICES

PROCESSOR

- Control Unit
  - is the component of the processor that directs and coordinates most of the operations in the computer.
- Arithmetic Logic Unit
  - performs arithmetic, comparison, and other operations.

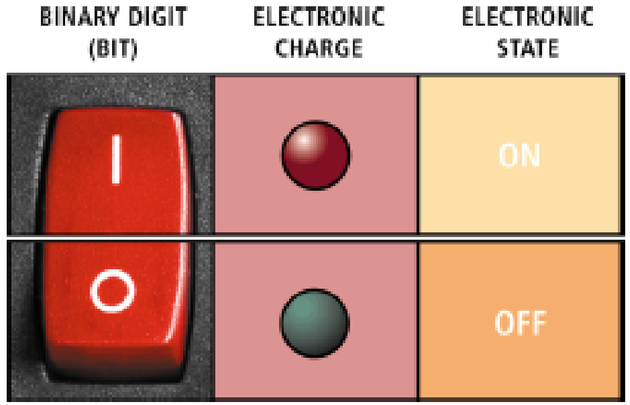
PROCESSOR

- For every instruction, a processor repeats a set of four basic operations, which comprise a machine cycle.



DATA REPRESENTATION

- Analog Signals
  - These are signals that are continuous and vary in strength and quality
    - Soundwaves
    - Thermometers
    - Vinyl records
- Digital Signals
  - These are signals that are discrete and represent information using binary values (0s and 1s)
    - CDs / MP3s
    - Digital Photo



MEMORY

- Memory consists of electronic components that store instructions waiting to be executed by the processor, data needed by those instructions, and the results of processing the data.
- It stores three (3) basic categories of items:
  - Application Programs
  - The Operating System and other System Software
  - Data being processed and the resulting information

MEMORY

Memory Sizes

Term	Abbreviation	Approximate Number of Bytes	Exact Number of Bytes	Approximate Number of Pages of Text
Kilobyte	KB or K	1 thousand	1,024	1/2
Megabyte	MB	1 million	1,048,576	500
Gigabyte	GB	1 billion	1,073,741,824	500,000
Terabyte	TB	1 trillion	1,099,511,627,776	500,000,000

- Petabyte (1,024 Terabytes or 1,048,576 gigabytes)
- Exabyte (1.024 Petabytes)
- Zettabyte (1,024 Exabytes)
- Yottabyte
  (1,024 Zettabytes or 1,208,925,819,614,629,174,706,176 bytes)

MEMORY

- The system unit has two (2) types of memory:
  - Volatile Memory
    - Loses its contents when the power is off
      - Example: Random Access Memory (RAM)
  - Nonvolatile Memory
    - Does not lose its contents when power is removed.
      - Examples:
        - Read-Only Memory (ROM)
        - Flash Drives

MEMORY

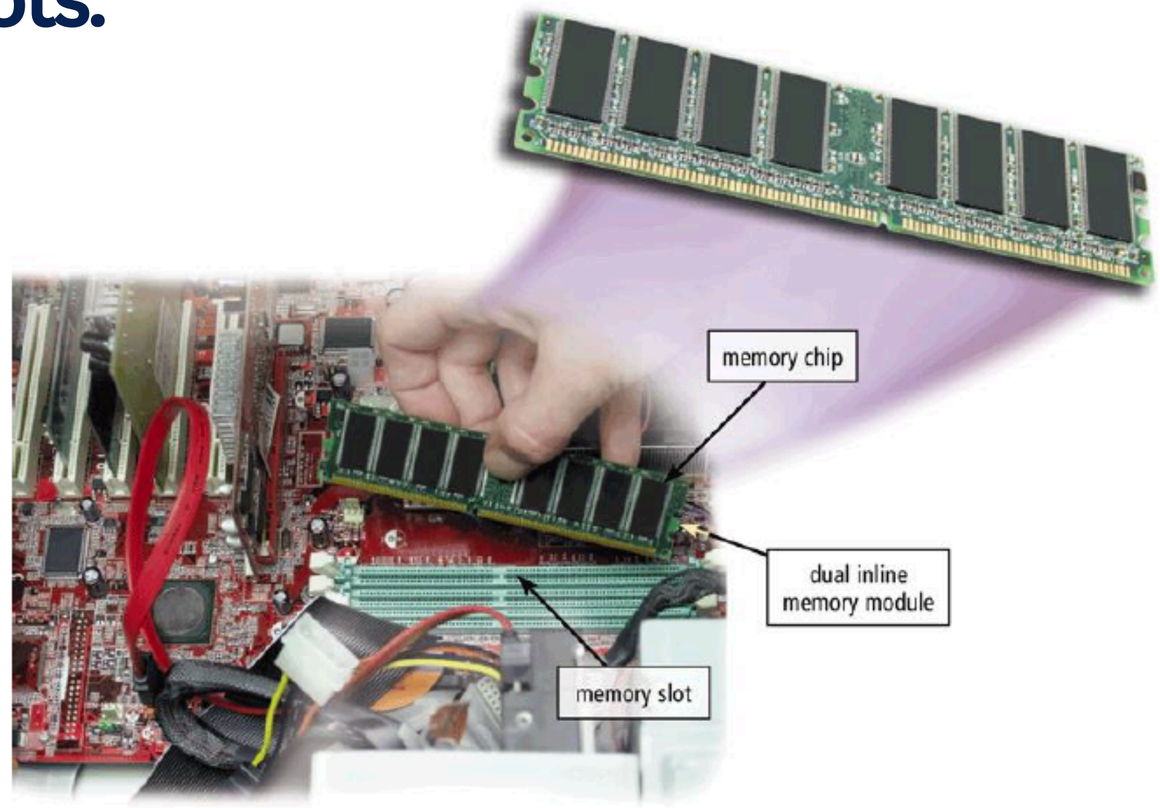
- Three (3) basic types of RAM chips:
  - Dynamic RAM (DRAM)
  - Static RAM (SRAM)
  - Magnetoresistive RAM (MRAM)

DRAM Variations

Name	Comments
SDRAM (Synchronous DRAM)	• synchronized to the system clock • much faster than DRAM
DDR SDRAM (Double Data Rate SDRAM)	• transfers data twice, instead of once, for each clock cycle • faster than SDRAM
DDR2	• second generation of DDR • faster than DDR
DDR3	• third generation of DDR • designed for computers with multi-core processors • faster than DDR2
RDRAM (Rambus DRAM)	• uses pipelining techniques • much faster than SDRAM

MEMORY

- RAM chips usually reside on a memory module and are inserted into memory slots.



MEMORY

- **The amount of RAM necessary in a computer often depends on the types of software you plan to use**

RAM	2 GB or less	2 GB to 8 GB	8 GB and up
Use	Home and business users managing personal finances; using standard application software such as word processing; using educational or entertainment optical discs; communicating with others on the Web	Users requiring more advanced multimedia capabilities; running number-intensive accounting, financial, or spreadsheet programs; using voice recognition; working with videos, music, and digital imaging; creating Web sites; participating in video conferences; playing Internet games	Power users creating professional Web sites; running sophisticated CAD, 3-D design, or other graphics-intensive software

MEMORY

- Read-Only Memory (ROM)
  - Refers to memory chips storing permanent data and instructions
- Programmable Read-Only Memory (PROM)
  - Is a blank ROM chip that can be written to permanently. Once programmed, the data cannot be changed.
- Flash Memory
  - can be erased electronically and rewritten

KEEPING YOUR COMPUTER OR MOBILE DEVICE CLEAN

- Clean your computer or mobile device once or twice a year.
- Turn off and unplug your computer or mobile device before cleaning it.
- Use compressed air to blow away dust
- Use an antistatic wipe to clean the exterior of the case and cleaning solution and soft cloth to clean the screen

INPUT AND OUTPUT DEVICES

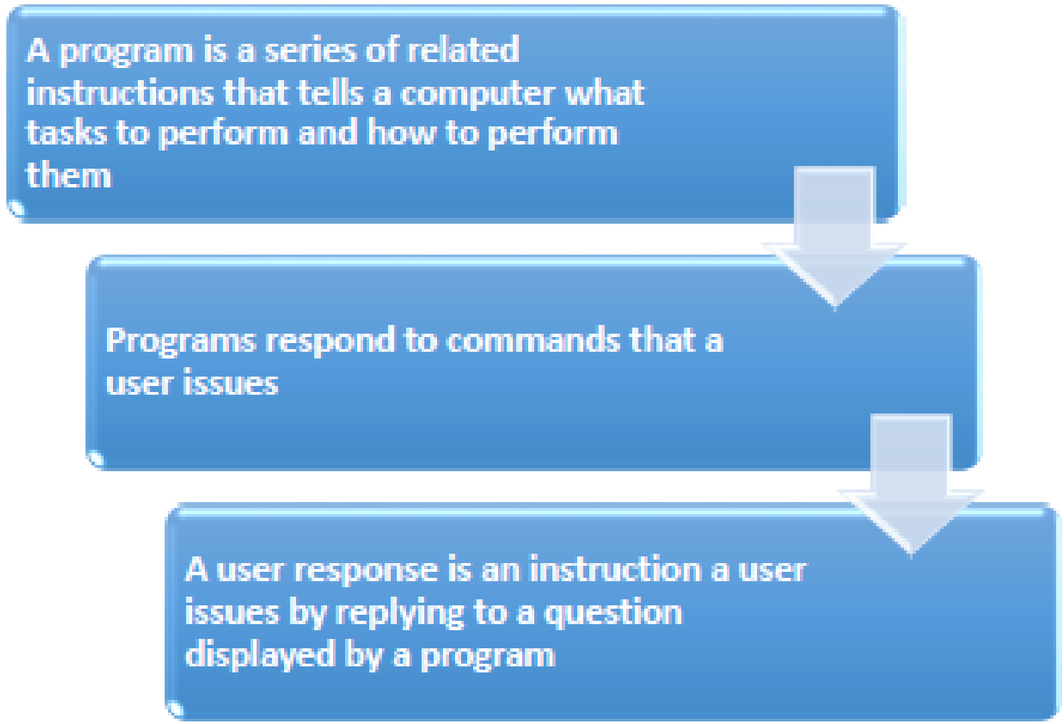
INPUT

- Input is any data and instructions entered into the memory of a computer.



WHAT IS INPUT?

- Instructions can be entered into the computer in the form of programs, commands, and user responses.



WHAT ARE INPUT DEVICES?

- An input device is any hardware component that allows users to enter data and instructions into a computer.











